SHAYINI SIDDARDHA

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PROFESSIONAL SUMMARY

AI/ML Engineer focused on local LLM inference optimization, autonomous AI agents, and computer vision. Creator of FastAgent, a hardware-adaptive inference orchestrator, and TAEL-Agent, a fine-tuned local SWE agent for resource-constrained systems. Experienced in LLM fine-tuning, agent runtime design, and open-source development, with research published in IEEE Xplore.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, HTML/CSS
AI/ML: LLMs, Agentic AI, LLM Fine-Tuning (QLoRA/LoRA), RAG, NLP, CV, scikit-learn, TensorFlow, YOLOv12n
LLM & Agent Systems: Ollama, llama.cpp, GGUF, KV-Cache Quantization, Prompt Caching, Grammar-Constrained Decoding, Tool Calling, Persistent Agent Memory
Frameworks & Tools: FastAPI, Flask, LangChain, LanceDB, SQLite, Git, GitHub, VS Code Extension Development, REST APIs

PROJECTS

FastAgent *(Open Source • Research Published)* | GitHub | Publication — Jun 2026 – Sep 2026

- Engineered a lightweight, hardware-adaptive LLM inference orchestrator using **llama.cpp**, LCP prompt caching, 8-bit KV-cache quantization, grammar-constrained decoding, and adaptive compute routing.
- Reduced steady-state prompt-evaluation latency from **~3,500 ms to ~73 ms** on a CPU-only, 8 GB RAM system, with rolling context management and safer tool execution.

TAEL-Agent *(Local Autonomous SWE Agent)* | GitHub | Hugging Face — Sep 2026 – Present

- Built a privacy-preserving SWE agent for CPU-only, 8 GB RAM systems, fine-tuning **IBM Granite 3.1 2B Instruct** using QLoRA on a 29,900-sample, three-phase curriculum and deploying a 4-bit GGUF model locally.
- Engineered an autonomous runtime with **9-category task routing, 10 sandboxed workspace tools, four-pillar persistent memory, and context-budget management**.
- Developed a VS Code extension with code generation, refactoring, diff previews, workspace-aware assistance, and autonomous tool execution; achieved **8/8 benchmark-domain passes** in the reported evaluation suite.

WildGuardAI *(IEEE Published)* | GitHub | Publication — Aug 2025 – Jun 2026

- Developed a real-time wildlife conflict monitoring system using **YOLOv12n, BoT-SORT tracking, Kalman filtering, Flask, and SQLite** to detect animals and generate proximity-based alerts.
- Evaluated the detection model at **91.2% mAP@0.5** and published the research in IEEE Xplore.

Open Source Contributor — sklearn-genetic-opt | GitHub PR #178 — Sep 2025 – Mar 2026

- Contributed documentation improvements for parallel processing in GASearchCV; merged PR #178 addressing issue #152.

WORK EXPERIENCE

AI Engineer Intern | Infosys Springboard — Feb 2025 – Apr 2025

- Developed an electricity demand and price forecasting system using Facebook Prophet with ~95% prediction accuracy.
- Built a Flask backend and interactive web interface for real-time forecasting.

AI Engineer Intern | YBI Foundation — Dec 2024 – Jun 2025

- Built an Air Quality Prediction System using Random Forest Regression on Indian city datasets.
- Achieved ~96% AQI category accuracy through optimized preprocessing and model training.

EDUCATION

Bachelor of Technology – Computer Science & Engineering (AI & ML) — 2022 – 2026
Vignan Institute of Technology and Science, Hyderabad | CGPA: 7.72/10.00

CERTIFICATIONS & ASSESSMENTS

- **Oracle:** Agentic AI Certified Foundations Associate (Jul 2026), Oracle Certified Foundations Associate (Oct 2025)
- **TCS iON NQT (IT):** Qualified – Dec 2025 | Strong performance in Python & problem-solving
- **NPTEL:** Programming with Generative AI – Nov 2025
- **Coursera:** Cloud Machine Learning Engineering & MLOps – Dec 2024
- **Infosys Springboard:** DSA, AI Primer, Principles of Generative AI, Generative Models for Developers